МИНИСТЕРСТВО НАУКИ И ВЫСШЕГО ОБРАЗОВАНИЯ РОССИЙСКОЙ ФЕДЕРАЦИИ

Федеральное государственное автономное образовательное учреждение высшего образования

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Отчет по лабораторной работе №3

По дисциплине “Администрирование Linux”

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Санкт-Петербург

2026

Задание

1. Выведите список всех подключенных репозитариев
2. Обновите локальные индексы пакетов в менеджере пакетов.
3. Выведите информацию о метапакете build-essential,
4. Установите метапакет build-essential, при этом определите какие компоненты будут установлены, а какие обновлены.
5. Найдите пакет, в описании которого присутствует строка «clone with a bastard algorithm»
6. Скачайте в отдельную директорию в домашнем каталоге архив с исходными кодами найденного в п.5 пакета.
7. Установите пакет из исходных кодов, скаченных в п.6
8. Если в конфигурационном файле пакета нет параметров установки пакета в систему, то добавьте их, так, чтобы пакет устанавливался в /usr/local/bin и на него назначались права — rwxr-xr-x. Проверьте выполнения этих директив.
9. Проверьте, что любой пользователь может запускать установленный пакет, но не тратьте на это более 5 минут.
10. Создайте файл task10.log, в который выведите список всех установленных пакетов.
11. Создайте файл task11.log, в который выведите список всех пакетов (зависимостей), необходимых для установки и работы компилятора gcc.
12. Создайте файл task12.log, в который выведите список всех пакетов (зависимостей), установка которых требует установленного пакета libgpm2.
13. Создайте каталог localrepo в домашнем каталоге пользователя root и скопируйте в него с сайта http://snapshot.debian.org/package/htop/ пять разных версий пакета htop. Это можно сделать с помощью wget или просто передав файлы на виртуальную машину используя протокол ssh и утилиту scp.
14. Сгенерируйте в каталоге репозитория файл Packages, который будет содержать информацию о доступных пакетах в репозитории и файл Создайте файл Release, содержащий описание репозитория. Например:

```
Origin: My Local Repo
Label: My Local Repo
Suite: stable
Version: 1.0
Codename: myrepo
Architectures: amd64
Components: main
Description: My local APT repository
```

15. Обновите кэш apt
16. Выведите список подключенных репозитариях и краткую информацию о них.
17. Создайте файл task16.log в который выведите список всех доступных версий htop
18. Установите предпоследнюю версию пакета
19. Скачайте из сетевого репозитория пакет nano. Пересоберите пакет таким образом, чтобы после его установки, появлялась возможность запустить редактор nano из любого каталога, введя команду newnano. Для работы с пакетом следует использовать dpkg-deb, а для установки dpkg. В файле протокола работы опишите использованные команды.
20. Бонусный вопрос с подвохом - что есть в APT?

Выполнение работы

Используемые команды

В ходе выполнения лабораторной работы использовались следующие команды:

- `apt` (update, install, remove, show, policy, source)
- `apt-cache` (policy, madison, depends, rdepends, show)
- `apt-get` (source)
- `dpkg` (-i)
- `dpkg-deb` (-x, -e, -b)
- `dpkg-query` (-l)
- `dpkg-scanpackages`
- `apt-ftparchive`
- `apt build-dep`
- `apt list`
- `xargs`

Задание 1

Вывел список всех подключенных репозиториев, включая основные и дополнительные.

```
[15-03-2026 15:45:08] {6e5cfd2a} student@vm-amd:~/lab-3$ cat /etc/apt/sources.list
# Ubuntu sources have moved to /etc/apt/sources.list.d/ubuntu.sources
[15-03-2026 15:45:23] {6e5cfd2a} student@vm-amd:~/lab-3$ cat /etc/apt/sources.list.d/ubuntu.sources
Types: deb
URIs: http://archive.ubuntu.com/ubuntu/
Suites: noble noble-updates noble-backports
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg

Types: deb
URIs: http://security.ubuntu.com/ubuntu/
Suites: noble-security
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg
```

```
[15-03-2026 15:45:32] {6e5cfd2a} student@vm-amd:~/lab-3$ ls /etc/apt/sources.list.d/ | xargs -I {} cat /etc/apt/sources.list.d/{}
Types: deb
URIs: http://archive.ubuntu.com/ubuntu/
Suites: noble noble-updates noble-backports
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg

Types: deb
URIs: http://security.ubuntu.com/ubuntu/
Suites: noble-security
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg
# See http://help.ubuntu.com/community/UpgradeNotes for how to upgrade to
# newer versions of the distribution.

## Ubuntu distribution repository
##
## The following settings can be adjusted to configure which packages to use from Ubuntu.
## Mirror your choices (except for URIs and Suites) in the security section below to
## ensure timely security updates.
##
## Types: Append deb-src to enable the fetching of source package.
## URIs: A URL to the repository (you may add multiple URLs)
## Suites: The following additional suites can be configured
##   <name>-updates   - Major bug fix updates produced after the final release of the
##                      distribution.
##   <name>-backports - software from this repository may not have been tested as
##                      extensively as that contained in the main release, although it includes
##                      newer versions of some applications which may provide useful features.
##                      Also, please note that software in backports WILL NOT receive any review
##                      or updates from the Ubuntu security team.
## Components: Aside from main, the following components can be added to the list
##   restricted  - Software that may not be under a free license, or protected by patents.
##   universe    - Community maintained packages. Software in this repository receives maintenance
##                 from volunteers in the Ubuntu community, or a 10 year security maintenance
##                 commitment from Canonical when an Ubuntu Pro subscription is attached.
##   multiverse  - Community maintained of restricted. Software from this repository is
##                 ENTIRELY UNSUPPORTED by the Ubuntu team, and may not be under a free
##                 licence. Please satisfy yourself as to your rights to use the software.
##                 Also, please note that software in multiverse WILL NOT receive any
##                 review or updates from the Ubuntu security team.
##
## See the sources.list(5) manual page for further settings.
Types: deb
URIs: http://archive.ubuntu.com/ubuntu/
Suites: noble noble-updates noble-backports
Components: main universe restricted multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg

## Ubuntu security updates. Aside from URIs and Suites,
## this should mirror your choices in the previous section.
Types: deb
URIs: http://security.ubuntu.com/ubuntu/
Suites: noble-security
```

Задание 2

Обновил локальные индексы пакетов, загрузив информацию из всех подключенных репозиториев.

```
[15-03-2026 15:46:42] {6e5cfd2a} student@vm-amd:~/lab-3$ sudo apt update
[sudo] password for student:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,538 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,835 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [246 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.5 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2,677 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [336 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [622 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.7 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2,813 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:17 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [544 B]
Get:18 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [976 kB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [219 kB]
Get:20 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:21 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [20.8 kB]
Get:22 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [6,732 B]
Get:23 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:24 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [650 kB]
Get:25 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:26 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 c-n-f Metadata [556 B]
Get:27 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,566 kB]
Get:28 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [319 kB]
Get:29 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:30 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [33.0 kB]
Get:31 http://archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [7,044 B]
Get:32 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:33 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [40.4 kB]
Get:34 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,368 B]
Get:35 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [368 B]
Get:36 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:37 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [30.7 kB]
Get:38 http://archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [18.2 kB]
Get:39 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [13.2 kB]
Get:40 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1,480 B]
Get:41 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Packages [780 B]
Get:42 http://archive.ubuntu.com/ubuntu noble-backports/multiverse Translation-en [372 B]
Get:43 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 15.0 MB in 3s (5,430 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
123 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Задание 3

Получил подробную информацию о метапакете build-essential, включая описание, зависимости и версию.

```
[15-03-2026 15:49:43] {6e5cfd2a} student@vm-amd:~/lab-3$ apt show build-essential
Package: build-essential
Version: 12.10ubuntu1
Priority: optional
Build-Essential: yes
Section: devel
Origin: Ubuntu
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Original-Maintainer: Matthias Klose <doko@debian.org>
Bugs: https://bugs.launchpad.net/ubuntu/+filebug
Installed-Size: 17.4 kB
Depends: libc6-dev | libc-dev, gcc (>= 4:12.3), g++ (>= 4:12.3), make, dpkg-dev (>= 1.17.11)
Task: xubuntu-desktop, ubuntustudio-photography, ubuntukylin-desktop, ubuntukylin-desktop, ubuntukylin-desktop-minimal, ubuntu-mate-core, ubuntu-mate-desktop, ubuntu-budgie-desktop-minimal, ubuntu-budgie-desk
top, ubuntu-budgie-desktop-raspi, ubuntu-unity-desktop, edubuntu-desktop-gnome-minimal, edubuntu-desktop-gnome, edubuntu-desktop-gnome-raspi, ubuntucinnamon-desktop-minimal, ubuntucinnamon-desktop
Download-Size: 4,928 B
APT-Manual-Installed: yes
APT-Sources: http://archive.ubuntu.com/ubuntu noble/main amd64 Packages
Description: Informational list of build-essential packages
 If you do not plan to build Debian packages, you don't need this
 package.  Starting with dpkg (>= 1.14.18) this package is required
 for building Debian packages.
 .
 This package contains an informational list of packages which are
 considered essential for building Debian packages.  This package also
 depends on the packages on that list, to make it easy to have the
 build-essential packages installed.
 .
 If you have this package installed, you only need to install whatever
 a package specifies as its build-time dependencies to build the
 package.  Conversely, if you are determining what your package needs
 to build-depend on, you can always leave out the packages this
 package depends on.
 .
 This package is NOT the definition of what packages are
 build-essential; the real definition is in the Debian Policy Manual.
 This package contains merely an informational list, which is all
 most people need.  However, if this package and the manual disagree,
 the manual is correct.
```

Задание 4

В процессе установки определил, что будут установлено ничего

```
[15-03-2026 15:50:28] {6e5cfd2a} student@vm-amd:~/lab-3$ sudo apt install --simulate build-essential
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
build-essential is already the newest version (12.10ubuntu1).
0 upgraded, 0 newly installed, 0 to remove and 123 not upgraded.
```

Задание 5

Найден пакет **bastet** (игра-тетрис) с описанием “ncurses Tetris clone with a bastard algorithm”.

```
[15-03-2026 19:04:04] {6e5cfd2a} student@vm-amd:~/lab-3/Downloads/nano_rebuild$ apt search . | grep -i "clone with a bastard algorithm"

WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

  ncurses Tetris clone with a bastard algorithm
```

Задание 6

Скачаны исходные коды пакета bastet в директорию `~/lab-3/Downloads/bastet-0.43/`.

```
[15-03-2026 16:28:46] {6e5cfd2a} student@vm-amd:~/lab-3/Downloads$ sudo apt-get source bastet
Reading package lists... Done
NOTICE: 'bastet' packaging is maintained in the 'Git' version control system at:
https://salsa.debian.org/games-team/bastet.git
Please use:
git clone https://salsa.debian.org/games-team/bastet.git
to retrieve the latest (possibly unreleased) updates to the package.
Skipping already downloaded file 'bastet_0.43-7build1.dsc'
Skipping already downloaded file 'bastet_0.43.orig.tar.gz'
Skipping already downloaded file 'bastet_0.43-7build1.debian.tar.xz'
Need to get 0 B of source archives.
Skipping unpack of already unpacked source in bastet-0.43
[15-03-2026 16:29:34] {6e5cfd2a} student@vm-amd:~/lab-3/Downloads$ ls -l
total 56
drwxr-xr-x 4 root root  4096 Mar 15 16:28 bastet-0.43
-rw-r--r-- 1 root root 12396 Dec 19  2023 bastet_0.43-7build1.debian.tar.xz
-rw-r--r-- 1 root root  1995 Dec 19  2023 bastet_0.43-7build1.dsc
-rw-r--r-- 1 root root 29115 May  6  2011 bastet_0.43.orig.tar.gz
[15-03-2026 16:29:37] {6e5cfd2a} student@vm-amd:~/lab-3/Downloads$
```

Задание 7

Пакет bastet успешно скомпилирован и установлен в систему.

```
[19-03-2026 22:57:42] {a654f287} student@vm-amd:~/lab-3/Downloads/bastet-0.43$ make clean
rm -f Ui.o main.o Block.o Well.o BlockPosition.o Config.o BlockChooser.o BastetBlockChooser.o bastet
[19-03-2026 22:58:24] {a654f287} student@vm-amd:~/lab-3/Downloads/bastet-0.43$ make
g++ -DNDEBUG -Wall   -c -o Ui.o Ui.cpp
Ui.cpp: In member function 'int Bastet::BorderedWindow::GetMinX()':
Ui.cpp:82:9: warning: variable 'y' set but not used [-Wunused-but-set-variable]
   82 |     int y,x;
      |         ^
Ui.cpp: In member function 'int Bastet::BorderedWindow::GetMaxX()':
Ui.cpp:94:9: warning: variable 'y' set but not used [-Wunused-but-set-variable]
   94 |     int y,x;
      |         ^
Ui.cpp: In function 'Bastet::Dot Bastet::BoundingRect(const std::string&)':
Ui.cpp:174:24: warning: narrowing conversion of '(maxlen + 1)' from 'size_t' {aka 'long unsigned int'} to 'int' [-Wnarrowing]
  174 |     return (Dot){maxlen+1,splits.size()};
      |                  ~~~~~~^~
Ui.cpp:174:38: warning: narrowing conversion of 'splits.std::vector<std::__cxx11::basic_string<char> >::size()' from 'std::vector<std::__cxx11::basic_string<char> >::size_type' {aka 'long unsigned int'} to '
nt' [-Wnarrowing]
  174 |     return (Dot){maxlen+1,splits.size()};
      |                           ~~~~~~~~~~~^~
Ui.cpp: In member function 'int Bastet::Ui::MenuDialog(const std::vector<std::__cxx11::basic_string<char> >&)':
Ui.cpp:236:17: warning: narrowing conversion of '(width + 5)' from 'size_t' {aka 'long unsigned int'} to 'int' [-Wnarrowing]
  236 |     Dot d={width+5,choices.size()};
      |            ~~~~~^~
Ui.cpp:236:32: warning: narrowing conversion of '(& choices)->std::vector<std::__cxx11::basic_string<char> >::size()' from 'std::vector<std::__cxx11::basic_string<char> >::size_type' {aka 'long unsigned int'
 to 'int' [-Wnarrowing]
  236 |     Dot d={width+5,choices.size()};
      |                    ~~~~~~~~~~~~^~
g++ -DNDEBUG -Wall   -c -o main.o main.cpp
g++ -DNDEBUG -Wall   -c -o Block.o Block.cpp
g++ -DNDEBUG -Wall   -c -o Well.o Well.cpp
g++ -DNDEBUG -Wall   -c -o BlockPosition.o BlockPosition.cpp
g++ -DNDEBUG -Wall   -c -o Config.o Config.cpp
g++ -DNDEBUG -Wall   -c -o BlockChooser.o BlockChooser.cpp
g++ -DNDEBUG -Wall   -c -o BastetBlockChooser.o BastetBlockChooser.cpp
BastetBlockChooser.cpp: In member function 'virtual Bastet::BlockType Bastet::BastetBlockChooser::GetNext(const Bastet::Well*, const Bastet::Queue&)':
BastetBlockChooser.cpp:118:76: warning: 'std::binder2nd<_Operation> std::bind2nd(const _Operation&, const _Tp&) [with _Operation = greater_equal<int>; _Tp = long int]' is deprecated: use 'std::bind' instead
-Wdeprecated-declarations]
  118 |     int pos=find_if(blockPercentages.begin(),blockPercentages.end(),bind2nd(greater_equal<int>(),random()%100)) - blockPercentages.begin();
      |                                                                            ~~~~~~~~^~~~~~~~~~~~~
```

Задание 8

В Makefile добавлена цель install с параметрами. После установки файл `/usr/local/bin/bastet` имеет права `-rwxr-xr-x` (755).

```
SOURCES=Ui.cpp main.cpp Block.cpp Well.cpp BlockPosition.cpp Config.cpp BlockChooser.cpp BastetBlockChooser.cpp
PROGNAME=bastet
LDFLAGS+=-lncurses
CXXFLAGS+=-ggdb -Wall
CXXFLAGS+=-DNDEBUG -Wall
CXXFLAGS+=-pg
LDFLAGS+=-pg

all: $(PROGNAME)

depend: *.hpp $(SOURCES)
	$(CXX) -MM $(SOURCES) > depend

include depend

$(PROGNAME): $(SOURCES:.cpp=.o)
	$(CXX) -ggdb -o $(PROGNAME) $(SOURCES:.cpp=.o) $(LDFLAGS) -lboost_program_options

install: $(PROGNAME)
	install -D -m 755 $(PROGNAME) $(DESTDIR)/usr/local/bin/$(PROGNAME)

clean:
	rm -f $(SOURCES:.cpp=.o) $(PROGNAME)

mrproper: clean
```

Задание 9

Пакет успешно запускается от любого пользователя. Тут видна ошибка из-за того, использую ssh + kitty

```
[15-03-2026 16:48:52] {6e5cfd2a} student@vm-amd:~$ bastet
bastet: using a user-specific high scores file: /home/student/.bastetscores
as the global high scores file /var/games/bastet.scores2 is not writable
[15-03-2026 16:49:07] {6e5cfd2a} student@vm-amd:~$ bastet
bastet: using a user-specific high scores file: /home/student/.bastetscores
as the global high scores file /var/games/bastet.scores2 is not writable
[15-03-2026 16:49:18] {6e5cfd2a} student@vm-amd:~$ sudo -u testuser bash
[15-03-2026 16:49:23] {6e5cfd2a} testuser@vm-amd:/home/student$ bastet
bastet: using a user-specific high scores file: /home/testuser/.bastetscores
as the global high scores file /var/games/bastet.scores2 is not writable
Error opening terminal: xterm-kitty.
```

Задание 10

Создан файл `task10.log` со списком всех установленных пакетов.

```
[15-03-2026 16:51:22] {6e5cfd2a} student@vm-amd:~/lab-3$ dpkg-query -l > ~/lab-3/task10.log
[15-03-2026 16:51:31] {6e5cfd2a} student@vm-amd:~/lab-3$ ls -l
total 116
drwxrwxr-x 3 student student   4096 Mar 15 16:28 Downloads
-rw-rw-r-- 1 student student     35 Mar 15 15:44 Makefile
-rw-rw-r-- 1 student student 108615 Mar 15 16:51 task10.log
[15-03-2026 16:51:34] {6e5cfd2a} student@vm-amd:~/lab-3$ cat task10.log
Desired=Unknown/Install/Remove/Purge/Hold
| Status=Not/Inst/Conf-files/Unpacked/half-conf/Half-inst/trig-aWait/Trig-pend
|/ Err?=(none)/Reinst-required (Status,Err: uppercase=bad)
||/ Name                                  Version                                  Architecture Description
+++-==================================-========================================-============-==============================================================================
ii  adduser                               3.137ubuntu1                             all          add and remove users and groups
ii  amd64-microcode                       3.20250311.1ubuntu0.24.04.1              amd64        Processor microcode firmware for AMD CPUs
ii  apparmor                              4.0.1really4.0.1-0ubuntu0.24.04.4        amd64        user-space parser utility for AppArmor
ii  apport                                2.28.1-0ubuntu3.8                        all          automatically generate crash reports for debugging
ii  apport-core-dump-handler              2.28.1-0ubuntu3.8                        all          Kernel core dump handler for Apport
ii  apport-symptoms                       0.25                                     all          symptom scripts for apport
ii  appstream                             1.0.2-1build6                            amd64        Software component metadata management
ii  apt                                   2.8.3                                    amd64        commandline package manager
ii  apt-utils                             2.8.3                                    amd64        package management related utility programs
ii  autoconf                              2.71-3                                   all          automatic configure script builder
ii  automake                              1:1.16.5-1.3ubuntu1                      all          Tool for generating GNU Standards-compliant Makefiles
ii  autopoint                             0.21-14ubuntu2                           all          tool for setting up gettext infrastructure in a source package
ii  autotools-dev                         20220109.1                               all          Update infrastructure for config.{guess,sub} files
ii  base-files                            13ubuntu10.3                             amd64        Debian base system miscellaneous files
ii  base-passwd                           3.6.3build1                              amd64        Debian base system master password and group files
ii  bash                                  5.2.21-2ubuntu4                          amd64        GNU Bourne Again SHell
```

Задание 11

Создан файл `task11.log` со списком всех зависимостей, необходимых для установки и работы компилятора gcc.

```
[15-03-2026 16:53:57] {6e5cfd2a} student@vm-amd:~/lab-3$ apt-cache depends gcc > ~/lab-3/task11.log
[15-03-2026 16:54:11] {6e5cfd2a} student@vm-amd:~/lab-3$ ls -l
total 120
drwxrwxr-x 3 student student   4096 Mar 15 16:53 Downloads
-rw-rw-r-- 1 student student     35 Mar 15 15:44 Makefile
-rw-rw-r-- 1 student student 108615 Mar 15 16:51 task10.log
-rw-rw-r-- 1 student student    400 Mar 15 16:54 task11.log
[15-03-2026 16:54:16] {6e5cfd2a} student@vm-amd:~/lab-3$ cat task11.log
gcc
  Depends: cpp
  Depends: cpp-x86-64-linux-gnu
  Depends: gcc-13
  Depends: gcc-x86-64-linux-gnu
  Conflicts: gcc-doc
 |Recommends: libc6-dev
  Recommends: <libc-dev>
    libc6-dev
  Suggests: gcc-multilib
  Suggests: make
    make-guile
  Suggests: manpages-dev
  Suggests: autoconf
  Suggests: automake
  Suggests: libtool
  Suggests: flex
  Suggests: bison
  Suggests: gdb
  Suggests: gcc-doc
```

Задание 12

Создан файл `task12.log` со списком пакетов, которые зависят от libgpm2.

```
[15-03-2026 16:54:22] {6e5cfd2a} student@vm-amd:~/lab-3$ apt-cache rdepends libgpm2 > ~/lab-3/task12.log
[15-03-2026 16:56:03] {6e5cfd2a} student@vm-amd:~/lab-3$ ls -l
total 124
drwxrwxr-x 3 student student   4096 Mar 15 16:53 Downloads
-rw-rw-r-- 1 student student     35 Mar 15 15:44 Makefile
-rw-rw-r-- 1 student student 108615 Mar 15 16:51 task10.log
-rw-rw-r-- 1 student student    400 Mar 15 16:54 task11.log
-rw-rw-r-- 1 student student    423 Mar 15 16:56 task12.log
[15-03-2026 16:56:06] {6e5cfd2a} student@vm-amd:~/lab-3$ cat task12.log
libgpm2
Reverse Depends:
  libgpm-dev
  vim-nox
  vim-motif
  vim-gtk3
  mc
  vim
  zhcon
  xwpe
  xemacs21-nomule
  xemacs21-mule-canna-wnn
  xemacs21-mule
  vim-nox
  vim-motif
  vim-gtk3
  twin
  mc
  links2
  links
  libt3widget2t64
  libnotcurses-core3
  libfinal0
  jed
  gpm
  fte-console
  emacs-pgtk
  emacs-nox
  emacs-lucid
  emacs-gtk
  elinks
  aumix
  w3m
  vim
  libncursesw6
  libncurses6
  brltty
  libaa1
```

Задание 13

Создан каталог `/root/localrepo` и загружены 5 разных версий пакета htop.

```
[15-03-2026 18:09:02] {6e5cfd2a} root@vm-amd:~/localrepo# ls -l
total 772
-rw-r--r-- 1 root root 126596 May  2  2021 htop_3.0.5-7_amd64.deb
-rw-r--r-- 1 root root 152412 Feb 27  2023 htop_3.2.2-2_amd64.deb
-rw-r--r-- 1 root root 163860 Feb 10  2025 htop_3.3.0-5_amd64.deb
-rw-r--r-- 1 root root 167776 Apr  8  2025 htop_3.4.0-5_amd64.deb
-rw-r--r-- 1 root root 171412 Jul 11  2025 htop_3.4.1-5_amd64.deb
```

Задание 14

Созданы файлы Packages и Release с информацией о локальном репозитории.

```
[15-03-2026 18:09:38] {6e5cfd2a} root@vm-amd:~/localrepo# sudo apt-ftparchive packages . > Packages
[15-03-2026 18:10:07] {6e5cfd2a} root@vm-amd:~/localrepo# ls -l
total 780
-rw-r--r-- 1 root root 126596 May  2  2021 htop_3.0.5-7_amd64.deb
-rw-r--r-- 1 root root 152412 Feb 27  2023 htop_3.2.2-2_amd64.deb
-rw-r--r-- 1 root root 163860 Feb 10  2025 htop_3.3.0-5_amd64.deb
-rw-r--r-- 1 root root 167776 Apr  8  2025 htop_3.4.0-5_amd64.deb
-rw-r--r-- 1 root root 171412 Jul 11  2025 htop_3.4.1-5_amd64.deb
-rw-r--r-- 1 root root   4829 Mar 15 18:10 Packages
```

```
[15-03-2026 18:10:11] {6e5cfd2a} root@vm-amd:~/localrepo# apt-ftparchive release . > Release
[15-03-2026 18:11:07] {6e5cfd2a} root@vm-amd:~/localrepo# ls -l
total 784
-rw-r--r-- 1 root root 126596 May  2  2021 htop_3.0.5-7_amd64.deb
-rw-r--r-- 1 root root 152412 Feb 27  2023 htop_3.2.2-2_amd64.deb
-rw-r--r-- 1 root root 163860 Feb 10  2025 htop_3.3.0-5_amd64.deb
-rw-r--r-- 1 root root 167776 Apr  8  2025 htop_3.4.0-5_amd64.deb
-rw-r--r-- 1 root root 171412 Jul 11  2025 htop_3.4.1-5_amd64.deb
-rw-r--r-- 1 root root   4829 Mar 15 18:10 Packages
-rw-r--r-- 1 root root    816 Mar 15 18:11 Release
```

```
Date: Sun, 15 Mar 2026 15:11:07 +0000
Origin: My Local Repo
Label: My Local Repo
Suite: stable
Version: 1.0
Codename: myrepo
Architectures: amd64
Components: main
Description: My local APT repository for lab work
MD5Sum:
 44d15e7a773241c27910da4c49eb3481             4829 Packages
 c516e1850c8dc086821b043dbd42a2d2               38 Release
SHA1:
 be656a9f7f7ea95e0bc09b6768bf556861013ead             4829 Packages
 897aa85b9ff71f2f2c1261d67ef9b5dbfc71550b               38 Release
SHA256:
 5c13d71235b7d6a11d85a0b80fbfda633790d83f8aa6670d0c80ff6885c8fdc2             4829 Packages
 2ff805c6272ae4c3df4d4985e6ddd06935a9611540d00941c0d715ff5dfb8a07               38 Release
SHA512:
 cd685bc6c37f1b5bc0354fa0806b7ff4de0429f2544f065706959075eb7563f5cc6f0d1e9fe6390795cf2f1f769004af5b29b67fb08856ce5c4167cb0d9b08c6             4829 Packages
 e84f6a01b4482c694d1dfcaae8c3558e8598f6fba6efe0f638b0e59a18825ae2a66b3d1b1a5f5b16e939d15501908ab967c0efee5be566a5d7383e4cfe1686d3               38 Release
```

Задание 15

Локальный репозиторий добавлен в источники, кэш apt успешно обновлён.

```
[15-03-2026 18:17:27] {6e5cfd2a} root@vm-amd:~/localrepo# cat /etc/apt/sources.list.d/localrepo.list
deb [trusted=yes] file:/root/localrepo ./
[15-03-2026 18:17:30] {6e5cfd2a} root@vm-amd:~/localrepo# apt update
Get:1 file:/root/localrepo ./ InRelease
Ign:1 file:/root/localrepo ./ InRelease
Get:2 file:/root/localrepo ./ Release [991 B]
Get:2 file:/root/localrepo ./ Release [991 B]
Get:3 file:/root/localrepo ./ Release.gpg
Ign:3 file:/root/localrepo ./ Release.gpg
Get:4 file:/root/localrepo ./ Packages [4,829 B]
Hit:5 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:6 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:7 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:8 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
53 packages can be upgraded. Run 'apt list --upgradable' to see them.
N: Download is performed unsandboxed as root as file '/root/localrepo/./InRelease' couldn't be accessed by user '_apt'. - pkgAcquire::Run (13: Permission denied)
```

Задание 16

Создан файл `task16.log` со списком всех подключенных репозиториев и краткой информацией о них.

```
[15-03-2026 18:17:36] {6e5cfd2a} root@vm-amd:~/localrepo# cat /etc/apt/sources.list
# Ubuntu sources have moved to /etc/apt/sources.list.d/ubuntu.sources
[15-03-2026 18:21:00] {6e5cfd2a} root@vm-amd:~/localrepo# ls -la /etc/apt/sources.list.d/
total 20
drwxr-xr-x 2 root root 4096 Mar 15 18:17 .
drwxr-xr-x 9 root root 4096 Mar 15 16:07 ..
-rw-r--r-- 1 root root   42 Mar 15 18:17 localrepo.list
-rw-r--r-- 1 root root  775 Mar 15 16:37 ubuntu.sources
-rw-r--r-- 1 root root 3060 Mar 15 16:37 ubuntu.sources.curtin.orig
[15-03-2026 18:21:20] {6e5cfd2a} root@vm-amd:~/localrepo# cat /etc/apt/
apt.conf.d/         auth.conf.d/        keyrings/           preferences.d/      preferences.d.save/ sources.list        sources.list.d/     trusted.gpg.d/
[15-03-2026 18:21:20] {6e5cfd2a} root@vm-amd:~/localrepo# cat /etc/apt/sources.list.d/localrepo.list
deb [trusted=yes] file:/root/localrepo ./
[15-03-2026 18:21:51] {6e5cfd2a} root@vm-amd:~/localrepo# cat /etc/apt/sources.list.d/ubuntu.sources
Types: deb
URIs: http://archive.ubuntu.com/ubuntu/
Suites: noble noble-updates noble-backports
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg

Types: deb
URIs: http://security.ubuntu.com/ubuntu/
Suites: noble-security
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg

Types: deb-src
URIs: http://archive.ubuntu.com/ubuntu/
Suites: noble noble-updates noble-backports
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg

Types: deb-src
URIs: http://security.ubuntu.com/ubuntu/
Suites: noble-security
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg
```


```
[15-03-2026 18:17:36] {6e5cfd2a} root@vm-amd:~/localrepo# cat /etc/apt/sources.list
# Ubuntu sources have moved to /etc/apt/sources.list.d/ubuntu.sources
[15-03-2026 18:21:00] {6e5cfd2a} root@vm-amd:~/localrepo# ls -la /etc/apt/sources.list.d/
total 20
drwxr-xr-x 2 root root 4096 Mar 15 18:17 .
drwxr-xr-x 9 root root 4096 Mar 15 16:07 ..
-rw-r--r-- 1 root root   42 Mar 15 18:17 localrepo.list
-rw-r--r-- 1 root root  775 Mar 15 16:37 ubuntu.sources
-rw-r--r-- 1 root root 3060 Mar 15 16:37 ubuntu.sources.curtin.orig
[15-03-2026 18:21:20] {6e5cfd2a} root@vm-amd:~/localrepo# cat /etc/apt/
apt.conf.d/         auth.conf.d/        keyrings/           preferences.d/      preferences.d.save/ sources.list        sources.list.d/     trusted.gpg.d/
[15-03-2026 18:21:20] {6e5cfd2a} root@vm-amd:~/localrepo# cat /etc/apt/sources.list.d/localrepo.list
deb [trusted=yes] file:/root/localrepo ./
[15-03-2026 18:21:51] {6e5cfd2a} root@vm-amd:~/localrepo# cat /etc/apt/sources.list.d/ubuntu.sources
Types: deb
URIs: http://archive.ubuntu.com/ubuntu/
Suites: noble noble-updates noble-backports
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg

Types: deb
URIs: http://security.ubuntu.com/ubuntu/
Suites: noble-security
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg

Types: deb-src
URIs: http://archive.ubuntu.com/ubuntu/
Suites: noble noble-updates noble-backports
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg

Types: deb-src
URIs: http://security.ubuntu.com/ubuntu/
Suites: noble-security
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg
```


Задание 17

Выведен список всех доступных версий пакета htop.

```
[15-03-2026 18:44:23] {6e5cfd2a} root@vm-amd:~/localrepo# apt list -a htop
Listing... Done
htop/stable 3.4.1-5 amd64 [upgradable from: 3.3.0-4build1]
htop/stable 3.4.0-5 amd64
htop/stable 3.3.0-5 amd64
htop/noble,now 3.3.0-4build1 amd64 [installed,upgradable to: 3.4.1-5]
htop/stable 3.2.2-2 amd64
htop/stable 3.0.5-7 amd64
```

Задание 18

Установлена предпоследняя версия пакета htop (3.4.0-5).

```
[15-03-2026 18:48:27] {6e5cfd2a} root@vm-amd:~/localrepo# apt install htop=3.4.0-5
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libnl-genl-3-200
Use 'sudo apt autoremove' to remove it.
Suggested packages:
  lm-sensors
The following packages will be upgraded:
  htop
1 upgraded, 0 newly installed, 0 to remove and 62 not upgraded.
Need to get 0 B/168 kB of archives.
After this operation, 4,096 B disk space will be freed.
```

```
[15-03-2026 18:48:37] {6e5cfd2a} root@vm-amd:~/localrepo# apt list -a htop
Listing... Done
htop/stable 3.4.1-5 amd64 [upgradable from: 3.4.0-5]
htop/stable,now 3.4.0-5 amd64 [installed,upgradable to: 3.4.1-5]
htop/stable 3.3.0-5 amd64
htop/noble 3.3.0-4build1 amd64
htop/stable 3.2.2-2 amd64
htop/stable 3.0.5-7 amd64
```

Задание 19

Создан модифицированный пакет newnano.deb с символической ссылкой. После установки команда newnano запускает редактор nano.

```
[15-03-2026 18:49:48] {6e5cfd2a} student@vm-amd:~/lab-3/Downloads$ apt download nano
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nano amd64 7.2-2ubuntu0.1 [282 kB]
Fetched 282 kB in 1s (335 kB/s)
[15-03-2026 18:50:01] {6e5cfd2a} student@vm-amd:~/lab-3/Downloads$ ls -l
total 336
drwxr-xr-x 4 student student   4096 Mar 15 16:45 bastet-0.43
-rw-r--r-- 1 root    root     12396 Dec 19  2023 bastet_0.43-7build1.debian.tar.xz
-rw-r--r-- 1 root    root      1995 Dec 19  2023 bastet_0.43-7build1.dsc
-rw-r--r-- 1 root    root     29115 May  6  2011 bastet_0.43.orig.tar.gz
-rw-r--r-- 1 student student 281778 Oct 15  2024 nano_7.2-2ubuntu0.1_amd64.deb
-rw-rw-r-- 1 student student    400 Mar 15 16:53 task11.log
```

```
[15-03-2026 18:55:09] {6e5cfd2a} student@vm-amd:~/lab-3/Downloads/nano_rebuild$ dpkg-deb -x ../nano_7.2-2ubuntu0.1_amd64.deb temp
[15-03-2026 18:55:30] {6e5cfd2a} student@vm-amd:~/lab-3/Downloads/nano_rebuild$ ls -l
total 4
drwxr-xr-x 4 student student 4096 Oct 10  2024 temp
[15-03-2026 18:55:36] {6e5cfd2a} student@vm-amd:~/lab-3/Downloads/nano_rebuild$ dpkg-deb -e ../nano_7.2-2ubuntu0.1_amd64.deb temp/DEBIAN
[15-03-2026 18:56:13] {6e5cfd2a} student@vm-amd:~/lab-3/Downloads/nano_rebuild$ ls -l
total 4
drwxr-xr-x 5 student student 4096 Mar 15 18:56 temp
[15-03-2026 18:56:15] {6e5cfd2a} student@vm-amd:~/lab-3/Downloads/nano_rebuild$ ls -l temp/
total 12
drwxr-xr-x 2 student student 4096 Oct 10  2024 DEBIAN
drwxr-xr-x 2 student student 4096 Oct 10  2024 etc
drwxr-xr-x 4 student student 4096 Oct 10  2024 usr
```

```
[15-03-2026 18:57:14] {6e5cfd2a} student@vm-amd:~/lab-3/Downloads/nano_rebuild/temp/usr/bin$ ln -s nano newnano
[15-03-2026 18:57:34] {6e5cfd2a} student@vm-amd:~/lab-3/Downloads/nano_rebuild/temp/usr/bin$ ls  -l
total 276
-rwxr-xr-x 1 student student 279040 Oct 10  2024 nano
lrwxrwxrwx 1 student student      4 Mar 15 18:57 newnano -> nano
lrwxrwxrwx 1 student student      4 Oct 10  2024 rnano -> nano
```

```
[15-03-2026 18:57:10] {6e5cfd2a} student@vm-amd:~/lab-3/Downloads/nano_rebuild/temp/usr/bin$ ls -l
total 276
-rwxr-xr-x 1 student student 279040 Oct 10  2024 nano
lrwxrwxrwx 1 student student      4 Oct 10  2024 rnano -> nano
[15-03-2026 18:57:14] {6e5cfd2a} student@vm-amd:~/lab-3/Downloads/nano_rebuild/temp/usr/bin$ ln -s nano newnano
[15-03-2026 18:57:34] {6e5cfd2a} student@vm-amd:~/lab-3/Downloads/nano_rebuild/temp/usr/bin$ ls  -l
total 276
-rwxr-xr-x 1 student student 279040 Oct 10  2024 nano
lrwxrwxrwx 1 student student      4 Mar 15 18:57 newnano -> nano
lrwxrwxrwx 1 student student      4 Oct 10  2024 rnano -> nano
[15-03-2026 18:57:41] {6e5cfd2a} student@vm-amd:~/lab-3/Downloads/nano_rebuild/temp/usr/bin$ cd
[15-03-2026 18:58:26] {6e5cfd2a} student@vm-amd:~$ cd lab-3/
[15-03-2026 18:58:28] {6e5cfd2a} student@vm-amd:~/lab-3$ cd Downloads/
[15-03-2026 18:58:32] {6e5cfd2a} student@vm-amd:~/lab-3/Downloads$ cd nano_rebuild/
[15-03-2026 18:58:41] {6e5cfd2a} student@vm-amd:~/lab-3/Downloads/nano_rebuild$ dpkg-deb -b temp newnano.deb
dpkg-deb: building package 'nano' in 'newnano.deb'.
[15-03-2026 19:02:35] {6e5cfd2a} student@vm-amd:~/lab-3/Downloads/nano_rebuild$ dpkg-deb -c newnano.deb | grep newnano
lrwxrwxrwx student/student      0 2026-03-15 18:57 ./usr/bin/newnano -> nano
```

```
[15-03-2026 19:02:59] {6e5cfd2a} student@vm-amd:~/lab-3/Downloads/nano_rebuild$ sudo dpkg -i newnano.deb
[sudo] password for student:
(Reading database ... 190889 files and directories currently installed.)
Preparing to unpack newnano.deb ...
Unpacking nano (7.2-2ubuntu0.1) over (7.2-2ubuntu0.1) ...
Setting up nano (7.2-2ubuntu0.1) ...
Processing triggers for install-info (7.1-3build2) ...
Processing triggers for man-db (2.12.0-4build2) ...
[15-03-2026 19:03:41] {6e5cfd2a} student@vm-amd:~/lab-3/Downloads/nano_rebuild$ which ne
needrestart          netaddr              netplan              netstat              networkd-dispatcher  newnano
neqn                 netcat               netqtop-bpfcc        networkctl           newgrp               newusers
[15-03-2026 19:03:41] {6e5cfd2a} student@vm-amd:~/lab-3/Downloads/nano_rebuild$ which newnano
/usr/bin/newnano
```

Задание 20

В apt есть:

- **apt** - основная команда для управления пакетами
- **apt-get** - классический инструмент управления пакетами
- **apt-cache** - работа с кэшем пакетов
- **apt-file** - поиск файлов в пакетах
- **apt-mark** - управление состоянием пакетов
- **apt-config** - работа с конфигурацией APT
- **apt-ftparchive** - создание файлов Packages и Release
- **apt-key** - управление GPG-ключами
- **apt-cdrom** - добавление CD/DVD репозиториев
- **apt-extracttemplates** - извлечение шаблонов для debconf
- **apt-sortpkgs** - сортировка индексных файлов пакетов

Ну если спрашивалось про структуру то вот:

- **Packages** - индексный файл с информацией о пакетах
- **Release** - файл с описанием репозитория
- **Release.gpg** - цифровая подпись Release
- **InRelease** - объединённый файл Release с подписью
- **Sources** - индексный файл с исходными кодами
- **Translation** - файлы локализации